Some Studies on Optical and Thermal Phenomena from the Perspective of Statistical Mechanics

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Abstract

Within the framework of statistical mechanics, this paper presents several studies concerning optical and thermal phenomena. This paper points out that optics and thermodynamics have long been regarded as independent fields of study, but from the perspective of statistical mechanics, optical phenomena and thermal phenomena are connected. Subsequently, this paper reconsiders the position and meaning of the thermal concept of temperature in statistical mechanics. On this basis, two phenomena respectively involving thermal and optical processes can be naturally explained: the phenomenon of system heating and the phenomenon of non-conservation of the total number of photons in a thermal-equilibrium radiation system. The description of these two phenomena are distinctly different from the usual dynamical description. In addition, within the framework of statistical mechanics, several seemingly different thermal phenomena can also be explained within a common statistical framework.

INTRODUCTION

Since the establishment of thermodynamics and statistical mechanics, heat has always been regarded as a mode of energy transfer, while optics has mainly concerned electromagnetic radiation and the quantization of energy. After Planck proposed the blackbody radiation law [1], Einstein, inspired by it, gave a correct explanation of the heat capacity of solids, thereby linking optical phenomena with thermal phenomena [2]. However, within the framework of statistical mechanics, the relation between optical and thermal phenomena has not yet been sufficiently clarified.

First, this paper proposes that, within the framework of statistical mechanics, there are two principal ways in which energy can raise the temperature of a system: one is the absorption of optical radiation involving optical processes, and the other is heat transfer involving thermal processes. These two ways are equivalent. This establishes certain connections between optical phenomena and thermal phenomena under statistical mechanics. Second, the concept of temperature is reconsidered in terms of its position in statistical mechanics. The phenomenon of the non-conservation of the total number of photons in a thermal-equilibrium radiation system is then explained from

a statistical perspective. Finally, the significance of choosing structure as the point of departure in statistical mechanics is discussed: it lies in the fact that several seemingly different thermal phenomena can be explained within a unified statistical framework.

The following presents several studies on optical and thermal phenomena from the perspective of statistical mechanics. This paper is divided into four parts.

The following discussion presents, from the perspective of statistical mechanics, a study of certain optical and thermal phenomena. This paper is divided into four parts.

In the first part, a theoretical framework and statistical derivation are first established. It is then proposed that the essence of an increase in the system temperature is an increase in the average number of quanta within the phase cells of the system. It is also pointed out that the two ways involving optical and thermal processes respectively are statistically equivalent. Finally, two concrete examples are used to further illustrate the statistical equivalence of these two ways.

The second part discusses the meaning of temperature in statistical mechanics. On the basis of the first part's claim concerning the essence of temperature increase, the second part raises the following question: what position does temperature occupy in statistical mechanics?

Temperature has sometimes been treated as a quantity to be determined in advance; in the fundamental thermodynamic relation $\frac{1}{T} = \frac{\partial S}{\partial E}$, temperature is expressed through the mathematical relation between entropy and internal energy, and here temperature is not treated as an a priori quantity. Later, with the development of statistical mechanics, temperature appears in the statistical description of energy. This paper argues that, from a statistical perspective that takes structure as its starting point, temperature is a statistical representation of the distribution of the system's energy within phase cells, and therefore has statistical meaning. On this basis, two phenomena can be described naturally: the phenomenon of system heating and the phenomenon of the nonconservation of the total number of photons in a thermal-equilibrium radiation system. The description of these two phenomena is clearly different from the usual dynamical description.

The third part further explains the phenomenon mentioned above, namely the nonconservation of the total number of photons in a thermal-equilibrium radiation system. On the basis of the first two parts, it explains the statistical meaning of photon number in a thermal-equilibrium radiation system. In addition, when the concept of the photon enters statistical mechanics, it loses its original meaning as a "particle" and instead appears as a statistical count.

In the fourth part, the basic starting points in statistical mechanics are divided into two types: particle ontology and structure ontology. A theory that takes particles as its starting point is here called particle ontology; a theory that takes structure as its starting point is here called structure ontology. The choice of different theoretical starting points leads to different routes in the explanation of physical phenomena, and this is manifested as different descriptions of the same physical phenomenon. The meaning of structure ontology is also discussed, and its significance lies in the possibility of placing several seemingly distinct thermodynamic phenomena within a unified statistical framework.

1 STATISTICAL MECHANICAL CONSIDERATIONS ON OPTICAL AND THERMAL PROCESSES

1.1 *Theoretical Framework and Statistical Derivation*

Phase space is divided into phase cells of size h^3. Bose proposed that the microscopic distribution on these phase cells determines the macroscopic properties of a radiation system [3, 4].

In a radiation system, there are quanta of different frequencies. The energy E^s carried by a quantum within the s-th phase cell is $h\nu^s$ (s ranges from 0 to ∞). If the volume of a phase cell is denoted by $dxdydzdp_xdp_ydp_z$, and its size is h^3, then, once the spatial range and momentum range are given, the number of phase cells in the system is [4]

$$\Gamma_{\text{tot}} = \frac{1}{h^3}\int dxdydzdp_xdp_ydp_z. \tag{1}$$

Now suppose that, within a unit energy interval, the number of phase cells is Δs, and that Δn quanta are distributed over them, where both Δs and Δn are very large numbers. We use $p_r\Delta s$ to denote the number of phase cells in Δs in which r quanta appear. p_r also denotes the probability that r quanta appear in the s-th phase cell; therefore, we use p_r^s to replace p_r. Clearly, for all phase cells s, the following condition is satisfied,

$$\sum_r p_r^s = 1. \tag{2}$$

Given p_r^s, the number of possible distributions of Δn among Δs phase cells is

$$\frac{\Delta s!}{\prod_{r=0}^{r=\infty}(p_r^s\Delta s)!}. \tag{3}$$

Here the range of r is from 0 to infinity. Taking the logarithm of the above expression and then using Eq.(2), we obtain

$$-\Delta s\sum_r p_r^s \ln p_r^s. \tag{4}$$

Therefore, for the entire radiation system, this expression may be written as a summation over all phase cells. Combining this with Boltzmann principle, one obtains the expression for the entropy S of the radiation system,

$$S = -k_{\text{B}}\sum_{s,r} p_r^s \ln p_r^s, \tag{5}$$

where k_{B} is Boltzmann constant.

When the radiation system reaches thermodynamic equilibrium, the entropy is required to attain its maximum value, while the total energy of the system is given by

$$\overline{E} = \sum_{s,r} E^s r p_r^s. \tag{6}$$

Here, $\sum_{s,r} rp_r^s$ is the average number of quanta in the s-th phase cell. We now seek the maximum value of Eq.(5) under the constraint of Eq.(6). Introducing the Lagrange multiplier β, and taking Eq.(2) into account, we obtain

$$p_r^s = (1 - e^{-\beta E^s})e^{-\beta E^s r}. \tag{7}$$

The expression for β is then determined. Substituting Eq.(7) into Eq.(5), we obtain

$$S = -k_B \sum_s \ln(1 - e^{-\beta E^s}) + k_B \beta \overline{E}. \tag{8}$$

Combining this with the thermodynamic relation $\frac{\partial S}{\partial \overline{E}} = \frac{1}{T}$, where T denotes the system temperature, gives $\beta = \frac{1}{k_B T}$. Therefore, combining Eq.(6) and Eq.(7), the average energy $\overline{E^s}$ in the s-th phase cell is

$$\overline{E^s} = \frac{E^s}{e^{E^s/k_B T} - 1}. \tag{9}$$

Replacing E^s by $h\nu^s$, we obtain the expression for the average energy of a Planck oscillator.

1.2 *Proposed Theoretical Perspective*

Below, consider a system in thermal equilibrium. The system temperature is related to the average number of quanta $\overline{n^s}$ corresponding to each phase cell. When the temperature of the system increases, $\overline{n^s}$ changes. For every increase of 1K, the average number of quanta corresponding to each phase cell increases by

$$\Delta\overline{n^s} = \frac{e^{\frac{E^s}{k_B T}}}{(e^{\frac{E^s}{k_B T}} - 1)^2} \frac{E^s}{k_B T^2}. \tag{10}$$

Eq.(10) indicates the essence of temperature increase: an increase in the average number of quanta within phase cells. In addition, from a macroscopic perspective, absorption of optical radiation and heat transfer are two main ways of increasing the temperature of a system. However, from the perspective of statistical mechanics, these two ways of heating the system are equivalent: both increase the system temperature by increasing the average number of quanta within phase cells. [5]

Therefore, in statistical mechanics, there is no essential distinction between these two ways of increasing temperature, which involve optical processes and thermal processes, respectively. They are two forms of manifestation under the same quantum statistical mechanism.

1.3 *Examples and Discussion*

1.3.1 *The Statistical Equivalence of the Two Ways in Vibrational Heat Capacity*

To further clarify the above theoretical viewpoint, the following takes the vibrational heat capacity of a diatomic molecule as an example to explain, within the framework of statistical mechanics, the statistical equivalence involving optical and thermal processes.

A diatomic molecule may be regarded as a one-dimensional linear harmonic oscillator formed by the relative vibration of two atoms. The average number of quanta contained in the i-th oscillator is expressed as

$$\overline{n_i} = \frac{1}{e^{\frac{\Theta_v}{T}} - 1}. \tag{11}$$

Here, $\Theta_v = E/k_B$, which is called the characteristic temperature of molecular vibration (it is assumed that the characteristic temperature is the same for any oscillator, and the energy carried by a oscillator is written as E). Taking the derivative with respect to temperature gives

$$\frac{\partial \overline{n_i}}{\partial T} = \frac{e^{\frac{\Theta_v}{T}}}{(e^{\frac{\Theta_v}{T}} - 1)^2} \frac{\Theta_v}{T^2}. \tag{12}$$

It can be seen that, when the system temperature is increased by 1K, the average number of quanta in each oscillator increases by the amount given in Eq.(12). This indicates the reason for the temperature increase of a diatomic molecular system: the increase in the average number of quanta within the phase cells of the system. It should also noted here that the increase in the average energy within the phase cells of the system comes from the energy transferred to the system by the external environment. From the perspective of statistical mechanics, whether this occurs through absorption of optical radiation or through heat transfer, the system's response to such external energy is the same in a statistical sense.

In addition, multiplying Eq.(12) first by E and then by N_A gives the increase in the vibrational energy of 1 mole of diatomic molecules. This increase is also the energy that the external environment needs to transfer to the system; and it is also similar to the molar vibrational heat capacity c_V^v given by Mr. Einstein [2]:

$$c_V^v = R \frac{e^{\frac{\Theta_v}{T}}}{(e^{\frac{\Theta_v}{T}} - 1)^2} (\frac{\Theta_v}{T})^2. \tag{13}$$

where $R = k_B N_A$.

Thus, it can be seen that, whether absorption of optical radiation or heat transfer is adopted, for the same diatomic molecular system, the energy required to raise its temperature by 1K is given by the vibrational heat capacity shown in Eq.(13). This shows the statistical equivalence of optical and thermal processes involved in vibrational heat capacity.

1.3.2 *The Statistical Equivalence of the Two Ways in Other Systems*

First, consider the statistical equivalence of the two ways of increasing the temperature of a black-body system. If energy enters the black-body through heat transfer, this energy will be absorbed by the black-body and redistributed among the phase cells. This way of increasing the system temperature is not essentially different from absorption of optical radiation; both increase the system temperature by increasing the number of quanta within the phase cells of the system.

Second, the two ways of increasing the temperature of metals or solids also have statistical equivalence. Similarly, within the framework of statistical mechanics, absorption of optical radiation and heat transfer both involve, at the microscopic level, the redistribution of the system's total energy and an increase in the number of quanta in its phase cells. Therefore, the two ways are statistically equivalent in raising the temperature of metals or solids.

2 THE MEANING OF TEMPERATURE IN STATISTICAL MECHANICS

On the basis of the theoretical framework and statistical derivation presented in the first section, it has been established that phase cells are taken here as the primary objects of study. The core issue is to examine the distribution of the system's energy over phase cells. In other words, by considering how the structure—composed of phase cells—is occupied by energy, one can describe the state of the system. Here, the phase cell is the basic unit of the structure. This section will show that the physical state of a system can be characterized by the distribution of energy over phase cells; in this sense, temperature has a definite statistical meaning.

Temperature was initially associated with the concept of heat, but with the development of thermodynamics the two gradually become separated. In thermodynamics, temperature is regarded as a parameter describing the macroscopic state of a system. What, then, is the position of temperature in statistical mechanics?

From the perspective of statistical mechanics, under the constraint of total energy alone, that is, by introducing only one Lagrange multiplier. And the distribution of energy among phase cells follows the principle of maximum entropy (i.e. the principle of maximizing the number of microscopic states), yielding the average energy $\overline{E^s}$ in a phase cell as [3–5]

$$\overline{E^s} = \frac{E^s}{e^{E^s/k_B T} - 1}. \tag{14}$$

where the superscript s denotes the s-th phase cell, E^s is the energy carried by an quantum within the phase cell, k_B is the Boltzmann constant, and T denotes the temperature of the system.

Now, dividing Eq.(9) by E^s, a dimensionless result can be obtained. We denote this result by $\overline{n^s}$, it denotes the average occupation number of quanta within the s-th phase cell, hereafter referred to simply as the occupation number. Its explicit expression is

$$\overline{n^s} = \frac{1}{e^{E^s/k_B T} - 1}. \tag{15}$$

It can be seen that the occupation number $\overline{n^s}$ is a statistical quantity. Moreover, when the occupation number changes, the temperature of the system changes as well. This may be understood as follows: the distribution of energy over phase cells and temperature are not in a causal relation, but rather in a relation of the represented and the representing. That is, temperature is a statistical representation of the distribution of the system's energy over phase cells. Put differently, the microscopic distribution of energy over the structure of the system is expressed macroscopically through the parameter of temperature.

In addition, the thermodynamic concept of temperature is determined through the partial derivative of entropy with respect to internal energy, $\frac{1}{T} = \frac{\partial S}{\partial E}$, which, in mathematical terms, establishes the relation among temperature, entropy, and internal energy. Statistical mechanics further enriches the physical content of this relation: temperature, as a macroscopic parameter, represents the distribution of the system's internal energy over phase cells.

In the traditional thermodynamic formulation, one may think the rise of the system's temperature is directly related to the energy transferred from the external environment. In statistical mechanics, however, this question needs to be reconsidered. The energy transferred from the outside does not directly affect the temperature of the system; rather, it changes the structure of the distribution of the system's energy over phase cells. This result is then characterized through the macroscopic parameter of temperature. More specifically, after external energy is transferred to the system, the system redistributes this energy among its phase cells (the distribution follows the principle of maximum entropy), temperature is the macroscopic manifestation of the average energy in each phase cell.

The Statistical Understanding of the Heating Process

On the basis of the above statistical positioning of temperature, the same physical phenomenon can be understood differently. The heating of a system is taken as an example below.

From the perspective of dynamics, since the system is acted upon by the external environment, the particles undergo transitions from lower energy levels to higher energy levels. During this process, the average energy of the particles increases, and macroscopically the system appears as an increase in temperature.

However, from the perspective of statistical mechanics, the focus is no longer on the transition behavior of individual particles, but on the distribution of the system's energy among phase cells. In this statistical sense, the reason for the rise in the system's temperature is the increase in the average energy within each phase cell; the macroscopic parameter of temperature represents this change.

Therefore, in statistical mechanics, the heating process more essentially reflects the redistribution of energy over phase cells. In this sense, the increase in the system's temperature is only a statistical manifestation.

3 This Statistical Meaning of Photon Number in a Thermal-Equilibrium Radiation System

In statistical physics, a thermal-equilibrium radiation system may be regarded as a photon energy field. From the perspective of the occupation of the phase cell structure by the total energy, it can be seen that, in a thermal-equilibrium radiation system, whether the total photon number of the system remains constant is not a question that needs to be judged. This is because photon number is a statistical quantity, and statistical quantities do not involve the question of conservation. This point will be explained in detail below.

First, as mentioned above, the occupation number is a statistical quantity. It is also a dimensionless numerical value, which only reflects the microscopic state of the system. Second, in physics, A single photon carries an energy E (for a photon, $E = h\nu$, where ν is the photon frequency). Combining these two points, one may conclude that the occupation number and photon number are equivalent in a statistical sense. Since the occupation number is a statistical quantity, photon number may likewise be understood as a statistical quantity.

In this way, when the concept of the photon enters statistical mechanics, it loses its original meaning as a "particle". Moreover, when photon number exists as a statistical quantity used to characterize the state of a system, whether the total photon number is conserved is therefore not a question that needs to be judged here. Nor should one ask where the photons have gone when the system's temperature changes; one can only ask how photon energy is redistributed among phase cells when system's temperature changes.

The Statistical Identity of the Photon

The above discussion has already shown that, in statistical mechanics, the number of photons is statistical quantity. Then, in a statistical sense, what exactly is the identity of the photon?

On the one hand, according to the above discussion, the number of photons is statistical quantity. On the other hand, in the expression for black-body radiation equilibrium, it is natural that no second Lagrange multiplier is introduced to constrain the total number of photons. Combining these two points, we may regard the photon as existing in statistics only as a count [6].

Extending from the viewpoint, several points can be obtained:

First, in this context, the light is no longer grasped a particle or as a wave in the usual physical sense, and the photon is interpreted as a count in statistics. Therefore, the photon no longer has the physical "existence" of a particle. Thus, there is no need to describe photons through a concrete physical picture; for example, there is no need to ask whether, in a thermal-equilibrium radiation system, the total number of photons is conserved. This is not a question that needs to be judged, because the photon itself is, in the structural description, a statistical manifestation of the distribution of the system's energy.

Second, in statistical mechanics that takes structure as its point of departure, the emphasis is placed on the importance of the overall structure of energy occupation rather than on the dynamical behavior of individuals. When the total energy within a system changes, it is first manifested as an adjustment of the structure of energy occupation, and on the macroscopic level, this change is represented through a change in temperature.

Finally, the questions considered from the structure as its point of departure don't involve individual details, such as interactions between photons or the current state of a single photon. It is concerned only with how energy is distributed over phase cells so that the entropy of the system reaches its maximum. Any macroscopic quantity is the result of statistics and does not involve the microscopic causal relations of individuals.

4 DIFFERENT PATHS INDICATED BY DIFFERENT POINTS OF DEPARTURE IN STATISTICAL MECHANICS: A DISCUSSION OF PARTICLE ONTOLOGY AND STRUCTURE ONTOLOGY

Section Overview. In the study of physical problems, one typically adopts a particular theoretical point of departure. In statistical physics, the present work classifies such points of departure into two types: particle ontology and structure ontology. Different choices of theoretical starting point lead to different paths of interpretation, which are reflected in differing descriptions of the same physical phenomena. The discussion that follows is organized along this line. At the same time, this work also considers the significance of structure ontology. Its significance lies in the possibility of placing several seemingly distinct thermodynamic phenomena within a unified statistical framework.

4.1 *The Origins of the Two Points of Departure*

THE ORIGIN OF PARTICLE ONTOLOGY. Early studies in physics proceeded from entities as their point of departure. For example, in mechanics, the motion of objects and the interactions between them are investigated. In thermodynamics, matter is regarded as composed of a large number of microscopic particles, and macroscopic thermodynamic phenomena are described on this basis. In atomic physics, the internal structure of atoms serves as the primary object of study, focusing on the motion of particles within atoms and their interactions. This includes, in statistical physics, the Boltzmann distribution, which is also formulated on the basis of a particle model. Theories that take particles as their point of departure are here referred to as particle ontology.

THE ORIGINS OF STRUCTURE ONTOLOGY. At present, two lines of reasoning can be identified that point toward taking overall structure as the primary object of study. One line arises from the uncertainty relations in quantum mechanics. In particular, the expressions for position uncertainty and momentum uncertainty imply, in a formal sense, that the product of the uncertainties in position and momentum at a given moment has a lower bound. When this bound is approached, the particle becomes completely blurred, and the concreteness of the particle appears to lose its significance. This aspect suggests that, when considering certain problems, the object of study may need to be shifted, rather than taking particles as the primary focus. Another line arises from Bose's method of statistical counting, through which the black-body radiation law is derived [3]. In this method, phase space is divided into cells of volume h^3, emphasizing the occupation of quanta within these cells. Here, the cells function as the basic units for describing structure in phase space. These two lines of reasoning indicate that it is possible to take structure as the primary object of study. Theories that take structure as their point of departure are here referred to as structure ontology [7,8].

4.2 *Different Paths Indicated by the Two Points of Departure*

In statistical physics, the Boltzmann distribution and the Bose distribution may be regarded as corresponding to two distinct paths arising from different points of depar-

ture. The former takes particle as its starting point, whereas the latter takes structure as its starting point. Owing to the difference in points of departure, the aspects of the problem on which they focus are also different. Particle ontology emphasizes dynamical issues such as the motion and interaction of particles. By contrast, structure ontology focuses on the distribution of system energy among phase cells(in this framework, only the conservation of total energy in an equilibrium system is taken as an independent constraint, correspondingly, only one Lagrange multiplier is introduced). For this reason, the two approaches differ markedly in their understanding of equilibrium state, temperature, and the problem of system heating.

First, they understand equilibrium state differently. In particle ontology, equilibrium state is often understood as a state gradually formed through the interaction of a large number of particles and through the traversal of various possible paths; therefore, it is more easily interpreted as a state reached through temporal evolution [9]. By comparison, the equilibrium state understood by structure ontology does not involve descriptions of temporal processes, interactions, or causal chains. Rather, it understands equilibrium state as the stable statistical distribution of energy among phase cells under the constraint of conserved total energy.

Second, they interpret the concept of temperature differently. Particle ontology holds that temperature reflects the motion state of microscopic particles, the higher the temperature of the system, the more intense the motion of the microscopic particles; the lower the temperature of the system, the less intense the motion of the microscopic particles. Structure ontology, by contrast, holds that temperature characterizes the distribution state of energy over phase cells, the greater the average number of quanta within each phase cell, the higher the system temperature appears; the smaller the average number of quanta within each phase cell, the lower the system temperature appears. In this sense, temperature is understood as the statistical representation of distribution of system energy over phase cells.

Third, the two approaches give different accounts of the phenomenon of system heating. From the perspective of particle ontology, when external energy is transferred into the system, particles move from lower energy levels to higher energy levels, in this process, the energy of particles increases. From the perspective of structure ontology, however, this process may be described as follows: external energy is transferred into the system, and the system energy is redistributed among phase cells, this distribution follows the principle of maximum entropy(i.e. the most probable macrostate corresponds to the largest number of microscopic configurations). As a result, the average number of quanta in each phase cell increases, which appears macroscopically as a rise in system temperature.

4.3 *The Relation Between the Two Points of Departure*

What needs to be clarified here is that, in statistical physics, the relation between particle ontology and structure ontology is not one of mutual negation, nor is there a simple distinction between right and wrong. Rather, the difference lies in the starting point chosen when studying a problem. Different starting points lead to different emphases

on the same problem, different methods of investigation, and, consequently, different conclusions.

Particle ontology takes particles as fundamental, focusing on their states of motion and interactions. When studying a problem, it often examines how, after a certain period of time, particles change from one state to another, and how the interactions between particles operate. This easily leads toward the path of dynamics. Therefore, according to the interpretation of particle ontology, equilibrium state is a state ultimately reached through a dynamical process of particles; temperature is also understood as a macroscopic manifestation of the intensity of microscopic particle motion; and the phenomenon of system heating is interpreted as an increase in the energy of the particles themselves.

Structure ontology is different. It does not first ask whether particles exist as entities, but instead considers how the energy of the system is distributed over phase cells, which are the smallest units constituting the structure. In statistical physics, structure ontology takes phase cells as fundamental. In this way, equilibrium state, temperature, and the problem of system heating are understood as different forms of expression of the distribution of energy over phase cells.

In summary, particle ontology focuses on particles–their dynamical processes, whereas structure ontology focuses on phase cells–the mode in which energy is distributed. Below, this paper discusses the significance of structure ontology. In some cases, structure ontology offers a more concise and direct interpretation of certain problems and can place these problems within a unified statistical framework.

4.4 *The Significance of Structure Ontology*

The uncertainty relation in quantum mechanics indirectly indicates that, when studying certain problems, the object of study should be shifted from particles to other objects. Bose's derivation of black-body radiation law by means of statistical methods further provided a new object of study: phase cells, which serve as the most basic units for describing structure. Thus, the door was opened to taking structure as the starting point.

In statistical mechanics, choosing structure ontology to describe problems can lead to concise and direct explanations in some cases. As noted above, structure ontology understands equilibrium state, temperature, and the phenomenon of temperature increase as different forms of expression of the distribution of energy over phase cells. This means that three seemingly different phenomena are placed within the same statistical framework: equilibrium state corresponds to a stable statistical distribution; temperature is a statistical representation of this distribution; and the phenomenon of temperature increase corresponds to a change in the statistical distribution.

Therefore, in statistical physics, structure ontology not only provides a new starting point for research questions and points toward an explanatory path different from that of particle ontology, but also provides a more concise and direct statistical description of thermodynamic phenomena and places these phenomena within a unified statistical framework.

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